Toan Ngo

Software Engineer

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Summary

Software Engineer with 4+ years of Go backend experience across B2B gaming, CRM, and SaaS products — specialising in microservice architecture, event-driven systems, and service migration. Established engineering standards across teams; built internal developer tooling that cut manual workflows from minutes to seconds. AWS Certified Solutions Architect – Associate.

Skills

Languages	Go (primary), JavaScript / TypeScript, SQL
Backend	gRPC, Protobuf, gRPC-Gateway, REST, JWT, OAuth2, Kafka, RabbitMQ
Databases	PostgreSQL, Redis, MongoDB, ArangoDB
Cloud & Infra	AWS (EC2, S3, EKS), GCP (GKE, Pub/Sub), Docker, Kubernetes
CI/CD & Obs.	Jenkins, ArgoCD, GitHub Actions, Helm, Jaeger, Prometheus
AI / Other	GitHub Copilot, OpenCode (Claude), React

Experience

Mercury Studio — HCMC, Vietnam

Software Engineer — Slotty (B2B gaming backend, 120+ Go microservices) — Jun 2025 – Present

- Migrated .NET/C# backend services to idiomatic Go across Edge, Integration, and Core layers; scaffolded new services where no Go counterpart existed; coordinated API contracts and gRPC-Gateway routing changes with the frontend team.
- Delivered full CRUD for a 3-tier person hierarchy (agents, players, sub-accounts) — create, update, suspend/disable, delete — with strict 1-level downline permission checks enforced across `integration/agent`, `integration/sub_account`, and `integration/supporter`.
- Implemented structured audit logging via `core/audit` covering all write operations, giving operators a complete, queryable trail of every downline action.
- Instrumented migrated services with Jaeger end-to-end tracing and Redis Cluster caching with a structured key naming convention, cutting repeated downstream gRPC calls on hot read paths.
- Maintained Jenkins CI/CD pipelines with parallel Docker builds and GitOps deployment via ArgoCD + Helm across dev, staging, and production environments.
- Built **MyTools**, an internal Go + React developer dashboard replacing manual DB/Redis queries for person lookup, API diffing, and cache cloning — compressing a ∼5-minute workflow to ∼5 seconds, adopted by the whole team.

GTG Software — HCMC, Vietnam

Software Engineer — Jan 2022 – Mar 2025

GTG CRM (Feb 2024 – Mar 2025)

- Architected the real-time messaging core of a HubSpot-alternative CRM from scratch using Go, gRPC, and WebSocket with OAuth2 (Keycloak); deployed to AWS EKS via GitHub Actions + ArgoCD with full observability (VictoriaMetrics, Loki, Jaeger).
- Designed the omni-channel messaging service as the platform's core feature, integrating across contact management, marketing, sales, and service hubs for a competitive launch.
- Established team-wide coding standards and architecture patterns for the microservice layer.
- Participated in Figma design reviews and aligned technical decisions with the CEO; documented architecture and workflows in Confluence to onboard new engineers.

Tokeet (Jun 2023 – Feb 2024)

- Re-wrote a legacy Perl messaging system in Go; introduced multi-channel conversation tracking, OAuth2 cloud integrations, and Amazon SQS fan-out for parallel processing.
- Built centralized conversation tracking with Go design patterns for maintainability; wrote deployment docs enabling non-technical project managers to manage the app on AWS.

Trydome (Jun 2022 – Jun 2023)

- Optimised critical PostgreSQL queries, integrated Stripe billing via REST API, and designed a FluxCD GitOps pipeline automating cloud database deployment for end customers.
- Cleaned and validated legacy customer data to improve reporting accuracy and overall data reliability across the platform.
- Collaborated on REST API design and infrastructure provisioning across AWS and GCP using FluxCD-based GitOps for automated customer database configuration management.

Internship (Jan 2022 – Jun 2022)

- Built and maintained Go microservices with PostgreSQL for business workflows — data modeling, CRUD implementation, input validation, and endpoint integration; implemented Kafka-based async processing with an idempotency mechanism to prevent duplicate consumption.

Education & Certifications

AWS Certified Solutions Architect – Associate (Credly) — April 2025

B.Sc. Computer Science · Ton Duc Thang University — 2021

English - B1+ | Vietnamese - Native